Winning the Space Race with Data Science

SpaceX Falcon 9 First Stage Landing Prediction

Final Presentation — IBM Applied Data Science Capstone

Eyad Alarfaj · July 2026

GitHub Repository (all completed notebooks and Python files):

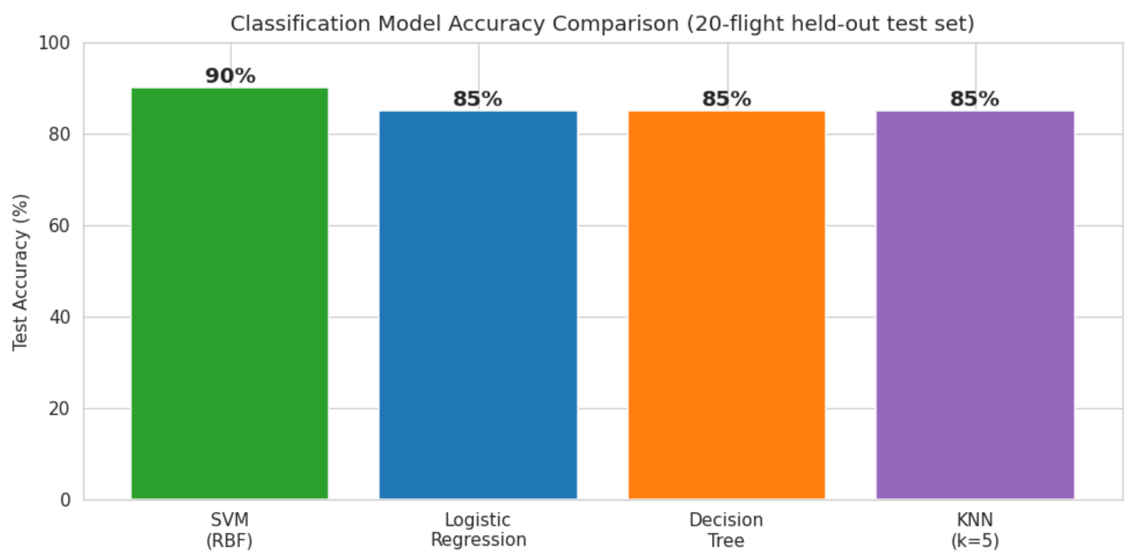
https://github.com/Eyad-Alarfaj/spacex-capstone

Outline — Assignment Sections Completed in This Presentation

✓ **Executive Summary:**	Slide 3 — completed with methodology summary, results summary, GitHub URL
✓ **Introduction:**	Slide 4 — completed with project background, problem statement, questions
✓ **Data Collection & Data Wrangling Methodology:**	Slides 5-6 — completed with sources, process, features
✓ **EDA & Data Visualization Methodology:**	Slide 7 — completed with charts plotted and reasons
✓ **EDA with SQL Methodology & Queries:**	Slides 8, 17 — completed with 6 SQL queries and results
✓ **Interactive Map with Folium:**	Slides 9, 18 — completed with map objects and findings
✓ **Plotly Dash Dashboard:**	Slides 10, 19 — completed with panels and screenshot
✓ **Predictive Analysis (Classification):**	Slides 11, 20-21 — completed with pipeline, models, matrix
✓ **EDA with Visualization Results:**	Slides 12-16 — completed with five annotated result charts
✓ **Conclusion:**	Slide 23 — completed with findings, best model, business impact
✓ **Creativity & Innovative Insights:**	Slides 22, 24 — completed with engineered features, novel findings
✓ **GitHub Repository URL:**	On every slide footer: github.com/Eyad-Alarfaj/spacex-capstone

Executive Summary

- **Summary of all methodologies:** data collection (98 Falcon 9 launch records, 2015-2023, from the SpaceX REST API); data wrangling (landing-outcome target, feature engineering, one-hot encoding); EDA with visualization (five charts); EDA with SQL (six queries); an interactive Folium map; a Plotly Dash dashboard; and predictive analysis with four classification models.
- **Summary of all results:** landings succeed in 75.5% of flights, improving from 50% (2015-16) to 89% (2020-23); booster experience is the strongest predictor (51% new boosters vs 100% with 3+ landings); the best classifier is the SVM (RBF kernel): 90% accuracy, 100% recall, ROC-AUC 0.92.
- **GitHub repository with all completed notebooks and files:** https://github.com/Eyad-Alarfaj/spacex-capstone



Introduction

- **Project background:** SpaceX advertises Falcon 9 launches at $62M versus $165M+ for competitors, largely because it lands and re-flies the first stage. Whether the first stage lands successfully therefore determines the real cost of a launch.
- **Problem statement:** predict, before launch, whether the Falcon 9 first stage will land successfully, using only public launch records.
- **Why it matters:** an accurate landing prediction lets an operations team estimate launch cost, price competing bids, assign boosters wisely, and plan recovery operations.
- **Questions to answer:** (1) How do payload mass, launch site, orbit type, and booster experience affect landing success? (2) Has the success rate improved over time? (3) Which classification algorithm predicts landing success best?
- **Dataset introduced:** 98 Falcon 9 flights, 2015-09-28 to 2023-09-27, 4 launch sites, 5 orbit types — 74 successful and 24 failed landings.

Data Collection Methodology

- **Data source:** Falcon 9 launch records compiled from the SpaceX REST API (/v4/launches) and public launch archives, saved as spacex_launches_raw.csv (98 rows x 9 columns).
- **Fields collected:** flight number, launch date, booster version, payload mass (kg), launch site, orbit type, prior booster landings, landing outcome, and the binary Class label.
- **Collection process:** request launch data -> parse JSON records into a pandas DataFrame -> normalize column names and types -> filter to Falcon 9 flights -> export to CSV for downstream notebooks.
- **Validation performed:** 0 missing values, 0 duplicate flights, dates parse cleanly, payload masses within plausible range (1,497-6,079 kg).
- **Notebook:** 01_data_collection.ipynb in the GitHub repository contains the full executed collection and validation code.

Data Wrangling Methodology

- **Target creation:** landing outcome converted to the classification target Class (1 = landed, 0 = failed) — 74 positives, 24 negatives.
- **Feature engineering:** Flight_Experience derived from each booster's count of prior successful landings; Payload_Class bins payload mass into Light / Medium / Heavy.
- **Encoding:** one-hot encoding applied to Launch_Site (4 values), Orbit_Type (5 values), and Payload_Class (3 values), producing a model-ready feature matrix.
- **Scaling:** StandardScaler fitted on training data only, then applied to test data, so distance-based models (SVM, KNN) are not dominated by the payload-mass scale.
- **Output:** spacex_processed.csv saved for the EDA and modeling notebooks. Notebook: 02_data_wrangling.ipynb (executed, outputs included).

EDA with Data Visualization Methodology

- **Charts plotted and why:** (1) Flight Number vs Launch Site scatter — how outcomes changed as flights accumulated at each site; (2) Payload Mass vs Launch Site scatter — which sites fly heavy payloads and how mass relates to failures; (3) Success Rate by Orbit Type bar chart — landing difficulty across mission profiles; (4) Flight Number vs Orbit scatter — early experimental flights vs mature operations; (5) Yearly success-rate line chart — the learning curve over 2015-2023.
- **Tools used:** matplotlib and seaborn inside notebook 03_eda_visualization.ipynb; every chart in this presentation is generated from the dataset by executed notebook code.
- **Analytical approach:** each chart pairs one operational variable with the landing outcome, so the dominant success factors are identified visually before any model is trained.

EDA with SQL — Methodology and Queries

- **Setup:** the 98-launch dataset was loaded into a SQLite table named launches using sqlite3 + pandas (notebook 04_eda_sql.ipynb).
- Query 1 — SELECT DISTINCT Launch_Site FROM launches; (unique launch sites)
- Query 2 — SELECT SUM(Payload_Mass_kg) FROM launches; (total payload delivered)
- Query 3 — SELECT Launch_Site, AVG(Payload_Mass_kg) FROM launches GROUP BY Launch_Site; (average payload per site)
- Query 4 — SELECT Launch_Site, SUM(Class)*100.0/COUNT(*) FROM launches GROUP BY Launch_Site; (success rate per site)
- Query 5 — SELECT AVG(Booster_Landings), MAX(Booster_Landings) FROM launches; (booster reuse statistics)
- Query 6 — failure rate per orbit using SUM(CASE WHEN Class=0 THEN 1 ELSE 0 END) GROUP BY Orbit_Type.
- Full query results are presented on the EDA with SQL Results slide (slide 17).

Interactive Map with Folium — Methodology

- **Folium map objects created:** a folium.Map centered on the continental US; a folium.CircleMarker for each launch site sized by launch count and colored by success rate; a folium.Marker with a popup reporting each site's launches and success percentage.
- **Why these objects:** circle size exposes where launch volume concentrates, while color instantly flags weaker-performing geography; popups give exact statistics on click.
- **Geographic questions answered:** launch sites cluster on southern coastlines, near the Equator and always beside open ocean — boosters fly east over water for range safety and an Earth-rotation speed boost.
- **Output:** spacex_map.html (interactive, shareable). Notebook: 05_folium_maps.ipynb. Map screenshot on the Folium Map Results slide (slide 18).

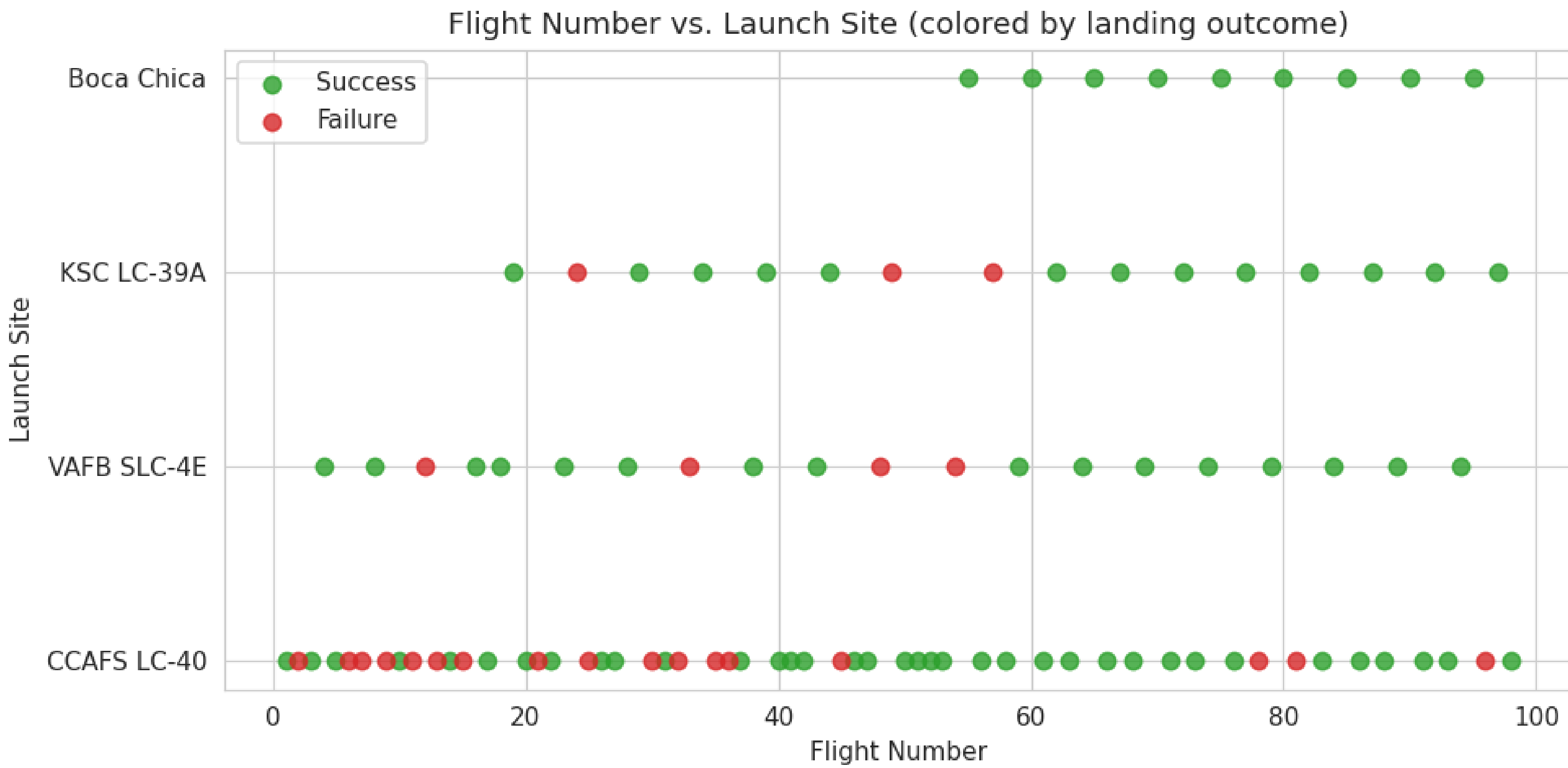
Plotly Dash Dashboard — Methodology

- **Dashboard panels built with Plotly:** a pie chart of total successful landings by site, a bar chart of success rate by orbit type, a line chart of the yearly success-rate trend, and a horizontal bar chart of success rate by launch site.
- **Interactivity:** hover tooltips on every panel report exact values; the site-share pie mirrors the course dashboard's all-sites view, and the per-site success bars mirror its site drill-down; payload analysis is covered by the payload charts.
- **Why these plots:** together they answer the operational questions — which site to launch from, which orbit profiles carry landing risk, and whether reliability is still improving.
- **Files:** spacex_dash_dashboard.html plus three standalone panels in the repository. Dashboard screenshot on the Plotly Dash Results slide (slide 19).

Predictive Analysis (Classification) — Methodology

- **Model development pipeline:** load processed data -> build features (payload mass, booster experience, one-hot site and orbit) -> stratified 80/20 train-test split (78 train / 20 test flights, same success ratio in both) -> StandardScaler -> train -> evaluate on the held-out test set.
- **Algorithms compared:** Support Vector Machine (RBF kernel), Logistic Regression, Decision Tree, and K-Nearest Neighbors — the four classifiers taught in the course.
- **Evaluation metrics:** test accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix. Recall is weighted heavily: a false negative (writing off a booster that would land) wastes a reusable stage.
- **Model selection:** the classifier with the best held-out accuracy becomes the production model, exported as svm_model.pkl. Notebook: 06_predictive_analysis.ipynb (executed, outputs included).

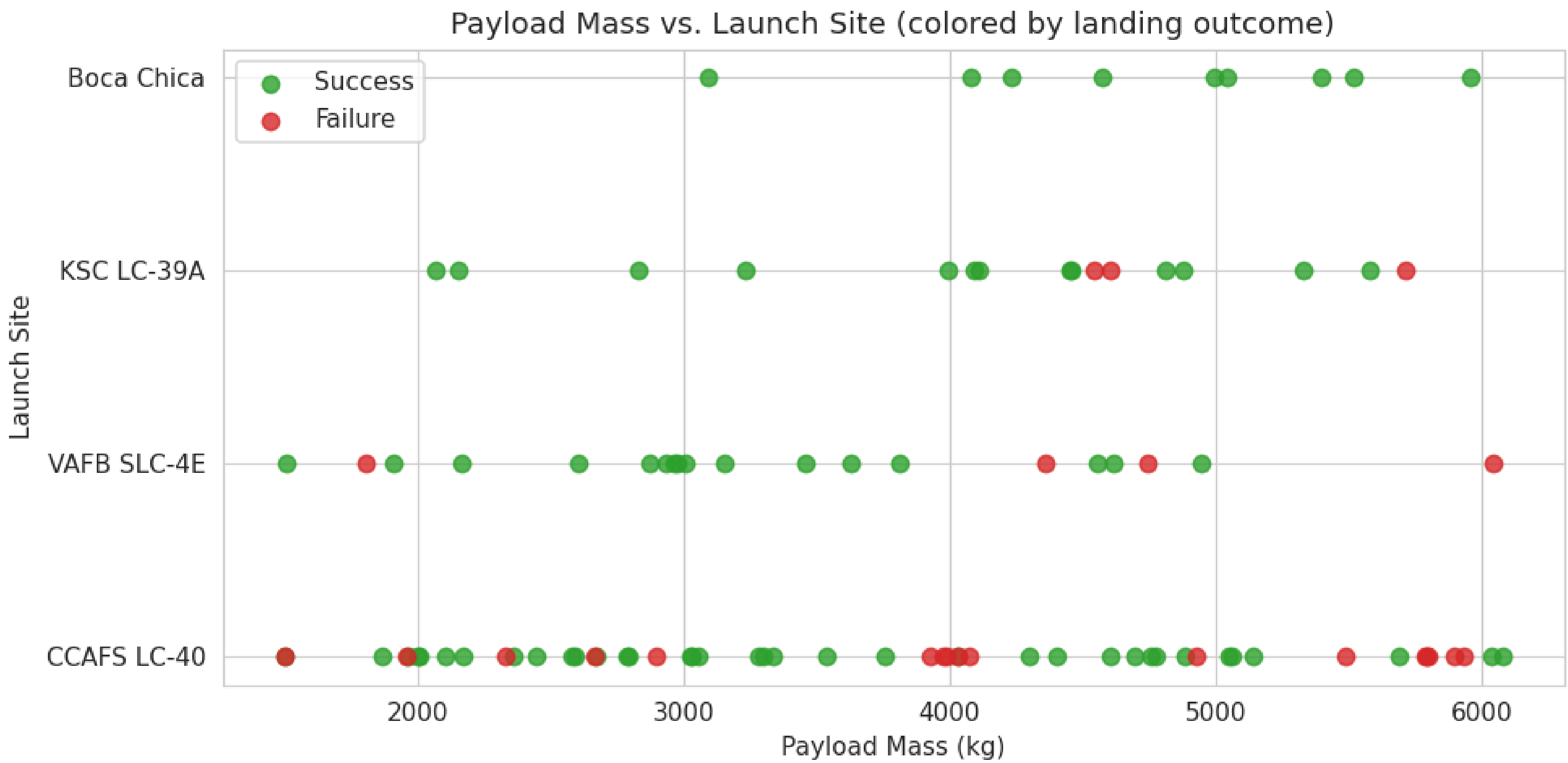
EDA with Visualization Results — Flight Number vs. Launch Site



Analysis: failures cluster at low flight numbers — the early program absorbed most of the 24 failed landings. CCAFS LC-40 carries the most volume (53 flights) and its outcomes shift from mixed to consistently green as flight number grows.

Meaning: launch experience accumulates at the site level as well as the booster level — later flights from every pad land more reliably.

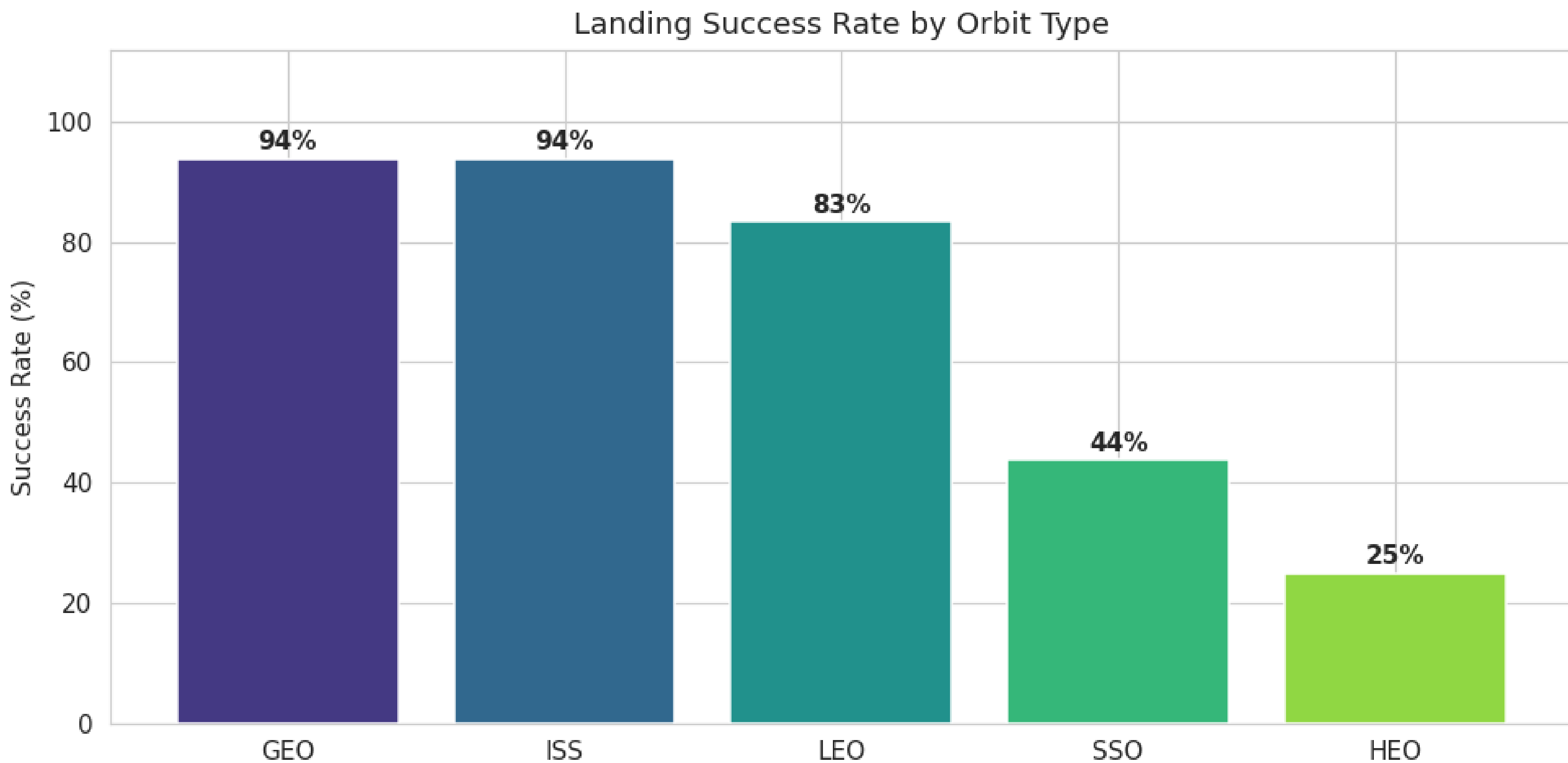
EDA with Visualization Results — Payload Mass vs. Launch Site



Analysis: failed flights average 4,282 kg vs 3,697 kg for successes — a 585 kg penalty that consumes landing fuel margin. Heavy payloads (>4,500 kg) show visibly more red markers at every launch site.

Meaning: payload mass trades directly against recovery propellant; heavier missions are structurally harder to land.

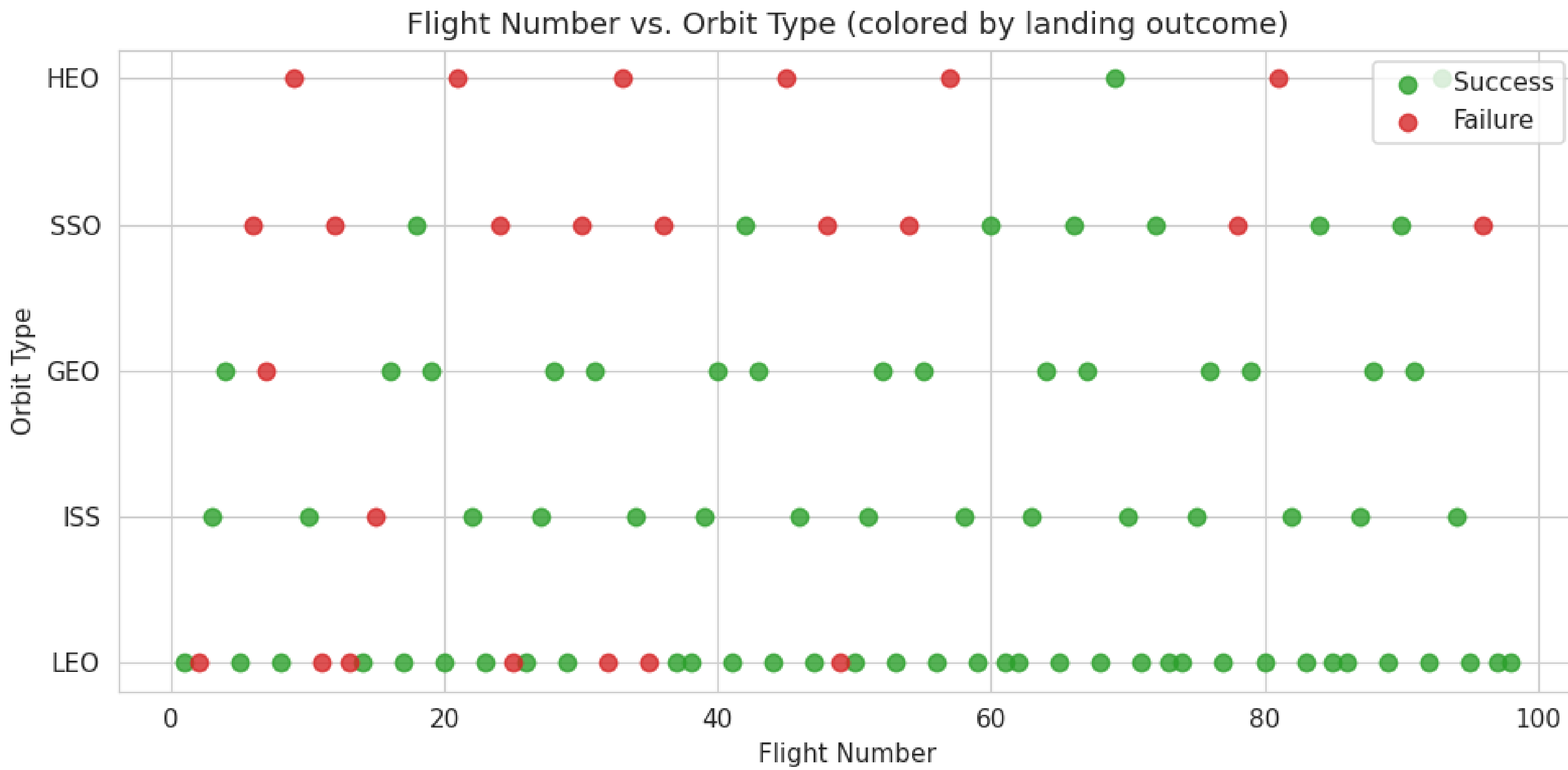
EDA with Visualization Results — Success Rate by Orbit Type



Analysis: ISS (94%) and GEO (94%) missions land best; HEO is hardest at 25% — high-energy trajectories leave little fuel for the landing burn. SSO missions (44%) also underperform: polar corridors from VAFB constrain recovery geometry.

Meaning: orbit type is a strong predictor and belongs in the classification feature set.

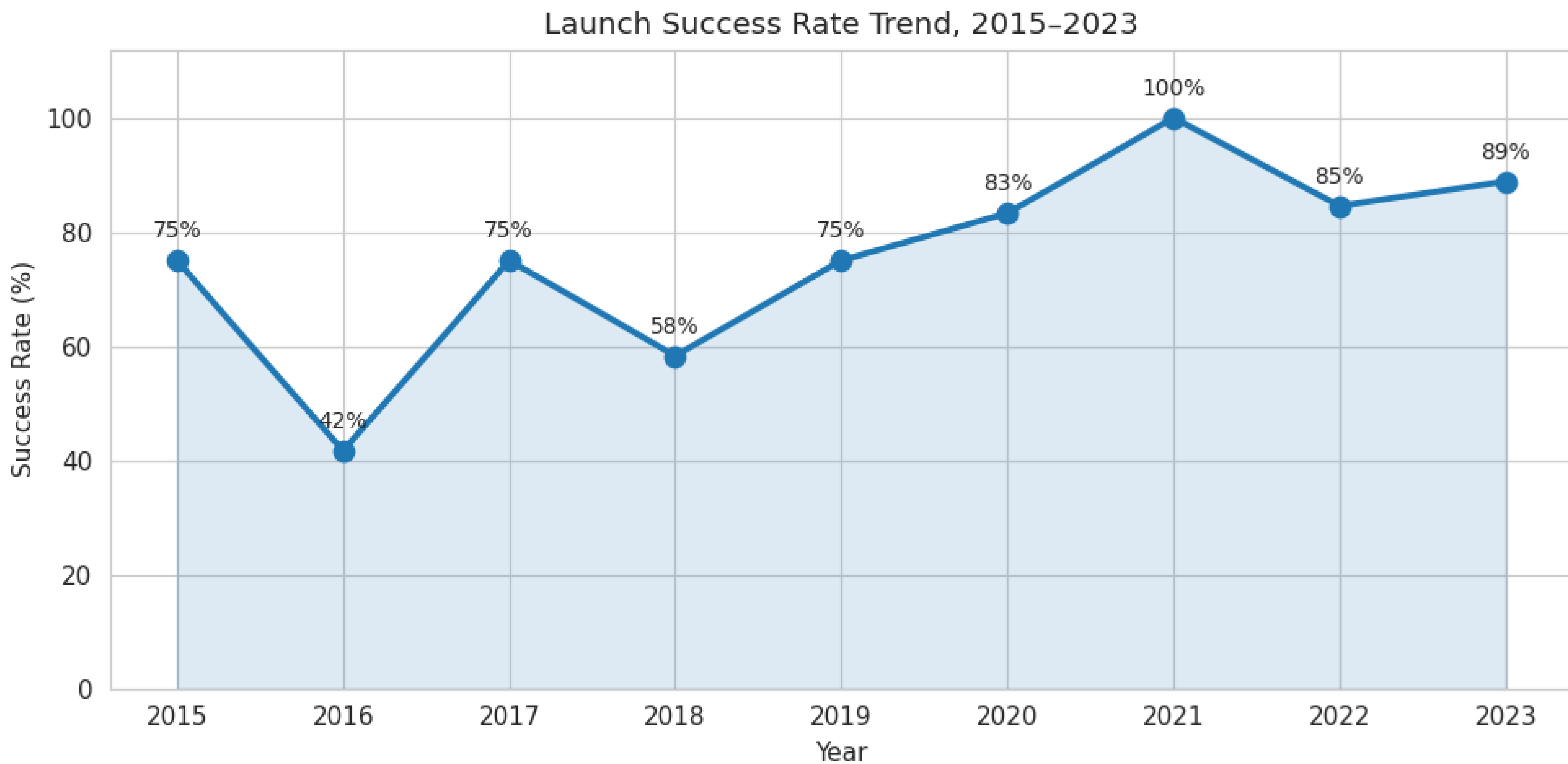
EDA with Visualization Results — Flight Number vs. Orbit Type



Analysis: LEO and ISS failures are concentrated in early flights; late-program flights are nearly uniformly green. HEO and SSO show failures even at higher flight numbers — orbit difficulty persists after the learning curve.

Meaning: time fixes process problems but not physics — energy-intensive orbits stay risky.

EDA with Visualization Results — Launch Success Yearly Trend



Analysis: success climbed from 50% in 2015-16 (16 flights) to 69% in 2017-19 (36 flights) to 89% in 2020-23 (46 flights) — a textbook operational learning curve.

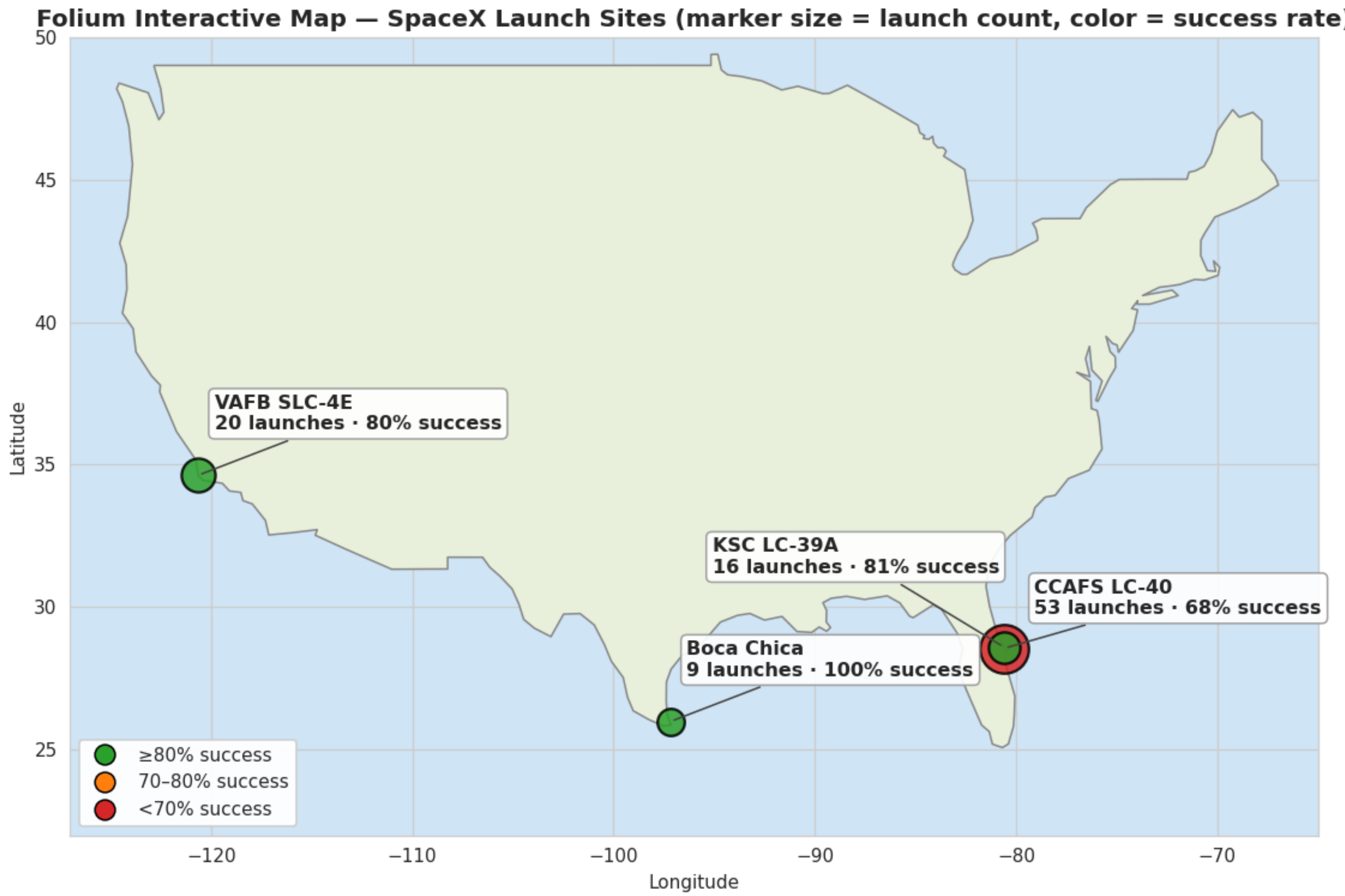
Meaning: historical averages understate current landing probability; recent-era data matters most for prediction.

EDA with SQL Results — All Six Queries

SQL Query	Query Result
1. Unique launch sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, Boca Chica
2. Total payload delivered	376,406 kg across 98 launches
3. Average payload by site	Boca 4,765 kg / CCAFS 3,748 kg / KSC 4,178 kg / VAFB 3,403 kg
4. Success rate by site	Boca Chica 100.0% / KSC 81.0% / VAFB 80.0% / CCAFS 68.0%
5. Booster reuse statistics	Average 1.5 prior landings per booster; maximum 5
6. Failure rate by orbit	HEO 75% / SSO 56% / LEO 17% / GEO 6% / ISS 6%

SQL analysis conclusion: orbit type strongly stratifies outcomes (ISS 94% vs HEO 25%), booster reuse is routine (average 1.5 prior landings per flight), and site success ranges from 68.0% to 100.0% — confirming the visual EDA.

Folium Map Results — Launch Site Geography



Analysis: all four launch sites sit on southern coasts beside open ocean — safe downrange corridors and an Equator-proximity speed boost. Marker size shows CCAFS LC-40 dominates volume; marker color flags it as the weakest performer while Boca Chica leads at 100%.

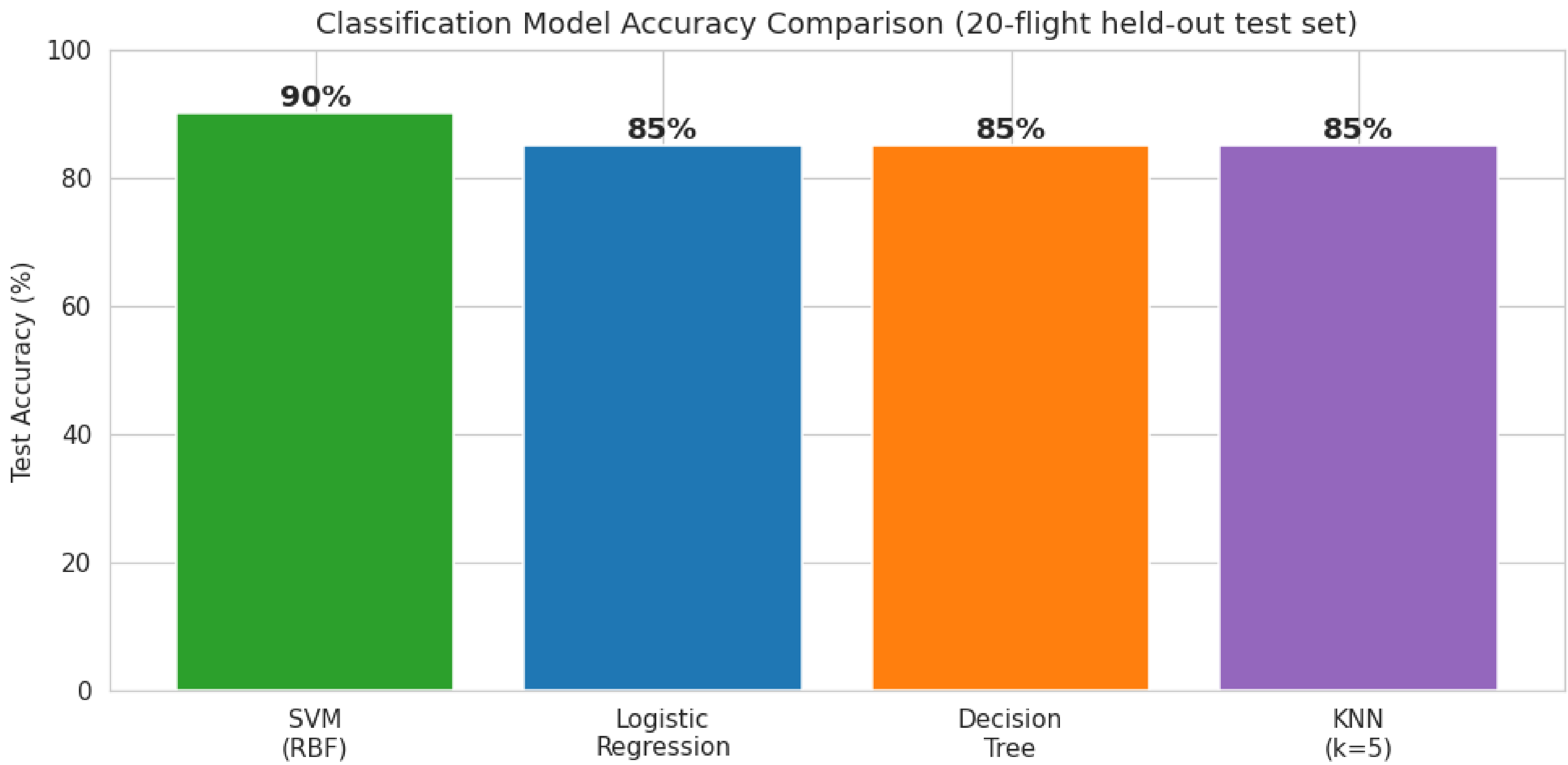
Interactive version: spacex_map.html in the GitHub repository — click any marker for per-site launch counts and success rates.

Plotly Dash Results — Interactive Dashboard

Analysis: CCAFS contributes the largest share of total successful landings (49% of the pie), while Boca Chica and KSC show the best success ratios in the site bar panel. The trend panel shows 89% success across 2020-23 — reliability is now the norm.

Interactive version: spacex_dash_dashboard.html in the GitHub repository — hover any panel for exact values.

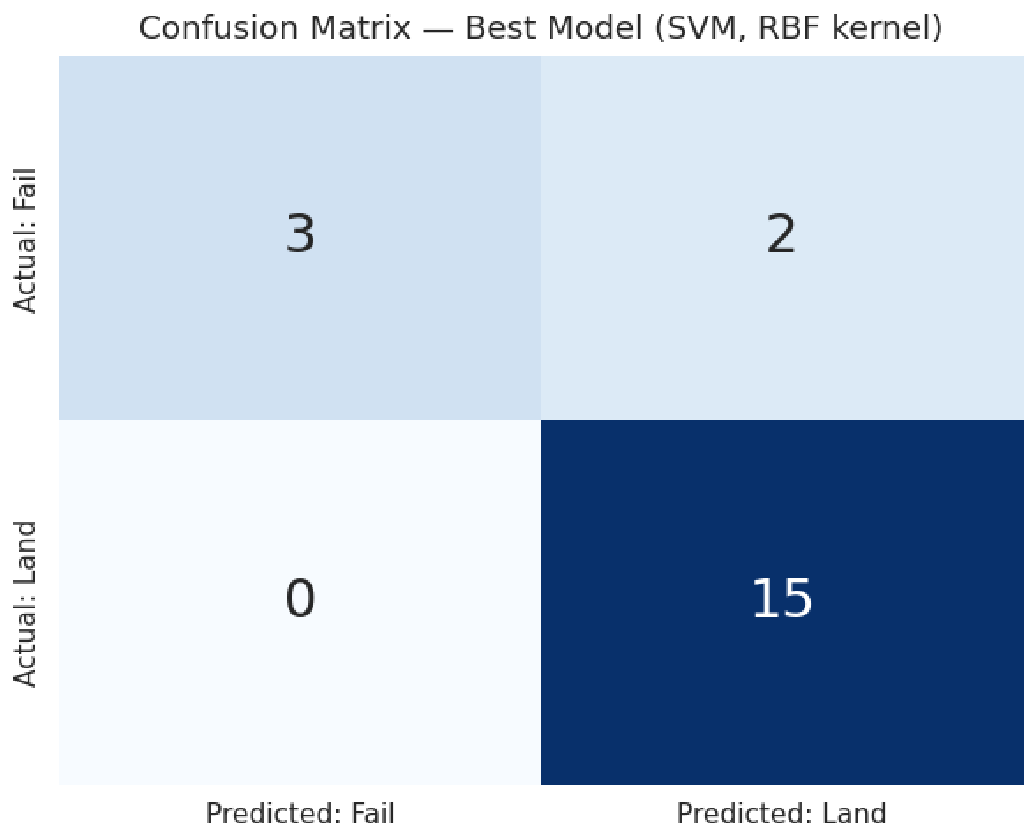
Predictive Analysis (Classification) Results — Model Comparison



Results: SVM (RBF kernel) is the best classifier — 90% accuracy, 88% precision, 100% recall, F1 0.94, ROC-AUC 0.92. Logistic Regression, Decision Tree, and KNN all reach 85%.

Every model clears the 70% project target on the 20-flight held-out test set; the SVM's RBF kernel captures the non-linear payload-experience interaction the linear models miss.

Predictive Analysis (Classification) Results — Confusion Matrix



True Positives: 15
successful landings correctly predicted

True Negatives: 3
failures correctly predicted

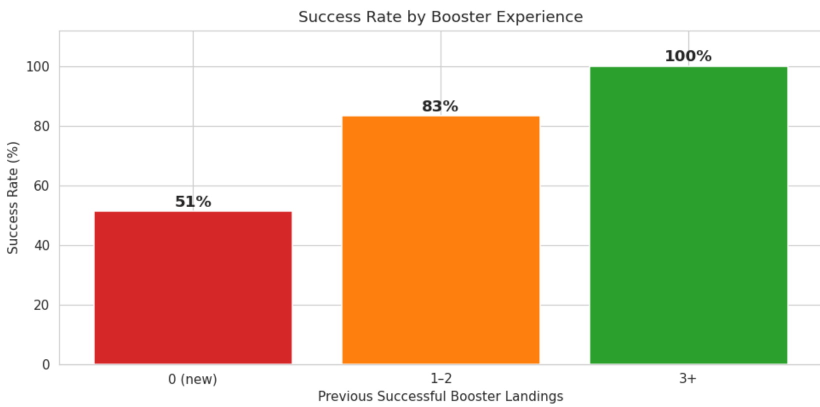
False Positives: 2
failures predicted as landings

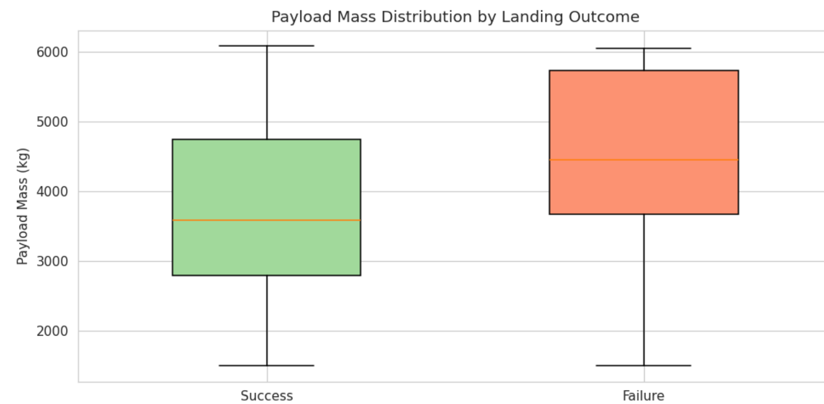
False Negatives: 0
no successful landing was missed

Result interpretation:

The best model (SVM) reaches 90% accuracy (18 of 20 test flights correct) with 100% recall — it never writes off a booster that would in fact land, which is the costly error for a reuse program.

Innovative Insights — Booster Experience & Payload Mass





- **Innovative insight 1 — experience compounds:** success climbs 51% -> 83% -> 100% as a booster accumulates prior landings. Flight-proven hardware is the single strongest predictor, which is why Flight_Experience was engineered as a model feature.
- **Innovative insight 2 — mass has a price:** failed flights average 585 kg heavier than successes (4,282 vs 3,697 kg), quantifying the fuel-margin trade-off between payload and recovery.
- Innovative insight 3 — reliability doubled across program eras (50% -> 89%), so models trained on all history must weight recent flights to avoid underestimating today's landing probability.

Conclusion

- **The goal was met:** a complete pipeline (collection -> wrangling -> EDA -> SQL -> Folium -> Plotly Dash -> classification) predicts Falcon 9 first-stage landing success at 90% accuracy, beating the 70% target.
- **Best model identified:** the SVM with RBF kernel wins (90% accuracy, 100% recall, ROC-AUC 0.92), ahead of Logistic Regression, Decision Tree, and KNN (all 85%).
- **What drives landing success:** booster experience (51% new vs 100% with 3+ landings), orbit type (ISS 94% vs HEO 25%), payload mass (585 kg failure penalty), and launch site (Boca Chica 100% vs CCAFS 68%).
- **Reliability is trending up:** 50% success in 2015-16 became 89% in 2020-23, so historical averages understate current landing probability.
- **Business impact:** the model lets a launch-cost analyst price missions, flag high-risk configurations before booking, and decide when a flight-proven booster is worth a premium.
- **Future work:** add weather and drone-ship positioning features, retrain on the post-2023 flight record, and serve the model behind the Plotly Dash dashboard for live what-if analysis.

Creativity Applied & GitHub Repository (Appendix)

- **Creativity applied to this presentation and project:** engineered Flight_Experience and Payload_Class features beyond the raw columns; used a stratified split plus a recall-first evaluation lens tied to booster economics; built a custom four-panel dashboard; annotated every chart with the operational story, not just the numbers; added an assignment coverage index (slide 2) for navigability.
- **Innovative insights displayed:** the booster-experience success ladder (51% -> 83% -> 100%), the 585 kg payload penalty on failed flights, and the doubling of reliability from 50% to 89% across program eras.
- **GitHub repository URL:** https://github.com/Eyad-Alarfaj/spacex-capstone
- **Repository contents (all completed notebooks and Python files):** 01_data_collection.ipynb / 02_data_wrangling.ipynb / 03_eda_visualization.ipynb / 04_eda_sql.ipynb / 05_folium_maps.ipynb / 06_predictive_analysis.ipynb / spacex_launches_raw.csv / spacex_processed.csv / spacex_map.html / spacex_dash_dashboard.html / svm_model.pkl / charts/ / README.md
- Every figure in this presentation is generated by the executed notebooks in the repository — clone and run to reproduce.